

UCE821 :HYDRAULIC STRUCTURES				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: To introduce the different types of dams and other dam related structures usually observed in a river				
Introduction: Types, classification, factors governing the selection of a dam site				
Gravity Dams: Forces acting on gravity dam, Design, Modes of failure and criteria for structural stability, principal and shear stress, methods of analysis, elementary profile of a gravity dam, high and low gravity dam, Joints in gravity dam, Galleries, foundation treatment				
Earth dam: Classification, Method of construction, Causes of failure, Design criteria, seepage line and its location, Design of Filters, Seepage control, Stability of earth dams.				
Spillways: Location, types, design consideration, energy dissipation below spillways, design of stilling basin, spillway crest gates.				
Coffer Dams: Use of cofferdams, types of coffer dams, merits and demerits, design of coffer dams.				
Surge tank: Function and types of surge tank, methods of surge analysis, design consideration of surge tank.				
River training works: Necessity, Causes of meandering, methods of training for rivers.				
Software Application: Settlement analysis, slope stability analysis of hydraulic structures under different conditions using Rocscience software.				
Course Learning Objectives (CLO)				
Upon completion of this course, the students will be able to:				
1. Design and analyse the various types of dams and surge tank. 2. Design the appropriate types of spillways. 3. Categorise the type of river training works				
Text Books				
1. Garg, S.K. ,Irrigation Engineering and Hydraulic structures, Khanna Publishers, New Delhi (2010). 2. Varshney, R.S., Concrete Dams, Oxford and IBH Publishing Company, New Delhi.(2009) 3. Novak, Pavel, Moffat, AIB, and Nalluri, C. ,Hydraulic structures, Taylor& Francis, (2007).				

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	30
2.	EST	45
3.	Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations)	25